SLIDE 1

Case Study: Optimizing Fruit Wholesale Revenue

- **Objective:** Maximize daily net profit for a Peruvian fruit wholesaler using a Linear Programming (LP) model.
- The analysis was conducted by first gathering real-world operational data—such as sales prices, costs, and storage limits—directly from the business owner to formulate a Linear Programming (LP) model focused on maximizing daily net revenue. This mathematical model defined specific decision variables for the quantity of each fruit type, caliber, and supplier, subject to constraints like warehouse capacity and quality standards . The problem was then solved using both LINGO software and Microsoft Excel's Solver function to identify the optimal purchasing strategy, with results cross-validated between the two tools to ensure accuracy. Finally, a sensitivity analysis was performed to evaluate the stability of the solution and determine how changes in key constraints, such as increasing storage capacity, would impact the overall profitability
- **Results:**
    - **Optimal Daily Revenue: S/3,793.75** (approx. ₹1.01 Lakh).
    - **Strategic Shift:** The model filled the warehouse (350/350 units) prioritizing high-margin items like **Kent Mangoes** and **Golden Pineapples**.
- **Key Recommendations:**
    - **Expand Storage:** Every extra unit of space adds **S/13.13** to daily profit (Shadow Price).
    - **Cut Product:** Discontinue **"Chato de Ica" Mango** due to low profitability.

SLIDE 2

key Constraints (Restrictions)

The business owner has several operational limits that the model must follow :

- **Storage Capacity:** Maximum of 350 fruit *jabas* (crates) in total.
- **Fruit Mix:** Sell equal or more pineapples than other fruits combined (mangoes and oranges).
- **Minimum Orange Sales:** At least 50 *jabas* of oranges.
- **Mango Ratio:** Sell twice as many Kent mangoes as Chato de Ica mangoes.
- **Average Caliber:** The overall average caliber of all fruits sold must be at least 'B'.
- **Minimum Caliber Volume:** A minimum of 100 *jabas* for each caliber (A, B, and C) must be sold.

- **Supplier Quotas:**
    - At least 25% of all products must come from the first supplier.
    - At least 40% of all products must come from the third supplier

SLIDE 3

1. Storage Capacity *(Total quantity must not exceed warehouse space)*

$$\sum_{i,j,k} x_{ij}^{k} \leq 350$$

2. Favored Fruit Strategy *(Pineapples must outnumber Mangoes and Oranges combined)*

$$\sum_{j,k} x_{\text{Pineapple},j}^{k} \geq \sum_{j,k} x_{\text{Mango},j}^{k} + \sum_{j,k} x_{\text{Orange},j}^{k}$$

3. Minimum Market Presence *(Must sell at least 50 crates of Oranges)*

$$\sum_{j,k} x_{\text{Orange},j}^{k} \geq 50$$

4. Mango Quality Ratio *(Kent Mangoes must be at least double Chato de Ica Mangoes)*

$$\sum_{j,k} x_{\text{Kent},j}^{k} \geq 2 \times \sum_{j,k} x_{\text{Chato},j}^{k}$$

5. Average Quality Standard *(Weighted average caliber must be at least 'B' / Value 2)*

$$\sum_{i,j,k} Q_j x_{ij}^{k} \geq 2 \sum_{i,j,k} x_{ij}^{k}$$

6. Minimum Variety (Caliber A) *(Must stock at least 100 crates of Caliber A)*

$$\sum_{i,k} x_{i,A}^{k} \geq 100$$

7. Minimum Variety (Caliber B) *(Must stock at least 100 crates of Caliber B)*

$$\sum_{i,k} x_{i,B}^{k} \geq 100$$

8. Minimum Variety (Caliber C) *(Must stock at least 100 crates of Caliber C)*

$$\sum_{i,k} x_{i,C}^{k} \geq 100$$

9. Supplier 1 Quota *(At least 25% of total stock from Supplier 1)*

$$\sum_{i,j} x_{ij}^{\text{Sup 1}} \geq 0.25 \sum_{i,j,k} x_{ij}^{k}$$

10. Supplier 3 Quota *(At least 40% of total stock from Supplier 3)*

$$\sum_{i,j} x_{ij}^{\text{Sup 3}} \geq 0.40 \sum_{i,j,k} x_{ij}^{k}$$

Slide 4

Our work